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Transradial Amputees – Proof of Concept Study**

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-Communication Protocol-

**Communication Protocol between the PC application
and the TR-FB device**

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1. Introduction

This document provides information about the Bluetooth communication protocol between the PC application and the TR-FB embedded system. A list of commands that may be sent to a TR-FB device as well as a list of data stream items that may be received from a TR-FB device once the connection between the PC application and the embedded system is established, will be given in the following document.

2. Communication parameters

Communication parameters are given in *Table 2.1*.

Table 2.1. Communication parameters.

Baud rate	Data bits	Parity	Stop bits	Handshake
115200	8	No	1	None

3. Commands from the PC application to the TR-FB device

A TR-FB embedded system will ignore wrong or incomplete messages. The commands that may be sent to a TR-FB device consist of n (*Command length*) bytes that are either ASCII characters or binary values and are given in the tables below.

The “*LEN*” parameter is calculated as follows:

$$LEN = Command\ length - 3$$

On the other hand, the “*CS*” parameter is calculated in the following way

$$CS = rem(sum(LEN, B1, B2, ..., Bn), 256),$$

where:

- B1 – the first byte after the “*LEN*” byte;
- B2 – the second byte after the “*LEN*” byte;
- Bn – the n-th byte after the “*LEN*” byte. It is also the last byte before the “*CS*” byte.

Table 3.1. Command 1 – Data stream control.

Command	> LEN S T x CS <
Decimal view	62 4 83 84 x CS 60
Hex view	0x3E 0x04 0x53 0x54 x CS 0x3C
Command length	7
Description	This command will either start or stop data stream from a TR-FB device, depending on the value of the “x” byte. The “x” byte can be either 1 (binary value) or 0 (binary value). By setting this byte to 1, the device will start sending appropriate data items to the PC application. On the other hand, by setting this byte to 0, the data stream will be stopped.

Table 3.2. Command 2 – Manual mode control.

Command	> LEN M x CS <
Decimal view	62 3 77 x CS 60
Hex view	0x3E 0x03 0x4D x CS 0x3C
Command length	6
Description	This command will either turn on or turn off manual control of the system, depending on the value of the “x” byte. The “x” byte can be either 1 (binary value) or 0 (binary value). By setting this byte to 1, manual mode will be turned on. On the other hand, by setting this byte to 0, manual mode will be turned off. If manual control of the system is turned on, it means that Michelangelo hand movements and vibration of the motors can be provided only by sending appropriate commands from the application. In other words, neither a hand nor vibration motors will be activated no matter how strong EMG signals are detected on two electrodes.

Table 3.3. Command 3 – Turning on/off vibration motors.

Command	> LEN V B W x CS <
Decimal view	62 5 86 66 87 x CS 60
Hex view	0x3E 0x05 0x56 0x42 0x57 x CS 0x3C
Command length	8
Description	This command will either turn on or turn off each of the eight vibration motors, depending on the value of the “x” byte. The “x” byte can be in the range of 0 (binary value) to 255 (binary value). This byte is calculated as follows: $x = \text{sum}([VB1, VB2, VB3, VB4, VB5, VB6, VB7, VB8]).*$ $[2^0, 2^1, 2^2, 2^3, 2^4, 2^5, 2^6, 2^7]), \text{ where } VBn = \begin{cases} 0, & \text{if } VBn \text{ is being turned off} \\ 1, & \text{if } VBn \text{ is being turned on} \end{cases}$

Table 3.4. Command 4 – Setting the duration of vibration of a certain motor.

Command	> LEN V B D x1 x2 CS <
Decimal view	62 6 86 66 68 x1 x2 CS 60
Hex view	0x3E 0x06 0x56 0x42 0x44 x1 x2 CS 0x3C
Command length	9
Description	This command will set the duration of vibration (the “x2” byte) of a certain motor (the “x1” byte). The “x1” byte can be in the range of 1 to 8 (binary value), while the “x2” byte can be in the range of 0 to 100 (binary value). By setting the “x2” to 0, vibration of a certain motor will be infinite. As for the values in the range of 1 to 100, a real vibration duration of a certain motor is obtained as follows: $\text{duration} = \frac{x2}{10} [s]$.

Table 3.5. Command 5 – Setting the vibration intensity of a certain motor.

Command	> LEN V B I x1 x2 CS <
Decimal view	62 6 86 66 73 x1 x2 CS 60
Hex view	0x3E 0x06 0x56 0x42 0x49 x1 x2 CS 0x3C
Command length	9
Description	This command will set the intensity of vibration (the “x2” byte) of a certain motor (the “x1” byte). The “x1” byte can be in the range of 1 to 8 (binary value), while the “x2” byte can be in the range of 1 to 100 (binary value). By setting the “x2” to 1, vibration intensity of a certain motor will be minimal (1%), while on

	the other hand, by setting this byte to 100, vibration intensity of a certain motor will be maximal (100%).
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Table 3.6. Command 6 – Manual move of the Michelangelo hand.

Command	> LEN C x1 x2_LB x2_HB CS <
Decimal view	62 5 67 x1 x2_LB x2_HB CS 60
Hex view	0x3E 0x05 0x43 x1 x2_LB x2_HB CS 0x3C
Command length	8
Description	This command will manually move the Michelangelo hand.
“x1” description	This byte (binary value) defines which movement will be performed: • 0 – hand stop • 1 – lateral close • 2 – pointed close • 3 – open pointed grip • 4 – lateral opening • 9 – pronation • 10 – supination
“x2” description	This parameter sets the speed of the movement and can be in the range of 1 to 1023, where: • “x2_LB” (binary value) – low byte of the movement speed • “x2_HB” (binary value) – high byte of the movement speed By setting this parameter to 0, manual control of the hand will be stopped. This command is equivalent to the <i>Command 2</i> , when turning manual mode off.

Table 3.7. Command 7 – EMG calibration and TR-FB hand control.

Command	> LEN M G x1 x2_LB x2_HB x3_LB x3_HB x4 x5 x6 x7 CS <
Decimal view	62 12 77 71 x1 x2_LB x2_HB x3_LB x3_HB x4 x5 x6 x7 CS 60
Hex view	0x3E 0x0C 0x4D 0x47 x1 x2_LB x2_HB x3_LB x3_HB x4 x5 x6 x7 CS 0x3C
Command length	15
Description	This command sends the calibrated EMG values either for CH1 (flexor) or CH2 (extensor), as well as the parameters on the basis of which TR-FB control of the Michelangelo hand is performed. If the command is received properly, the TR-FB device will respond with the message consisting of ASCII characters (“>OK<”).
“x1” description	This byte defines the channel from which the EMG signal is calibrated. It can be either 1 (binary value) or 2 (binary value), depending on whether the signal from the first (CH1) or the second channel (CH2) is calibrated.
“x2” description	This parameter defines the lower limit of the calibrated EMG range (when a muscle is relaxed) and consists of two bytes: • “x2_LB” (binary value) – low byte • “x2_HB” (binary value) – high byte For further information, see the section 5 in the document “Deliverable D1.2”.
“x3” description	This parameter defines the upper limit of the calibrated EMG range (when a muscle is maximally contracted) and consists of two bytes: • “x3_LB” (binary value) – low byte • “x3_HB” (binary value) – high byte For further information, see the section 5 in the document “Deliverable D1.2”.

"x4" description	This byte is the <i>"Grip – Rotation threshold"</i>. For further information, see the section 5.1 in the document "Deliverable D1.2". It can be in the range of 1 to 100 (binary value). The default value is 60. By setting this parameter to 1, the threshold is at the 1% of the calculated EMG range ["x2", "x3"], while on the other hand, by setting this parameter to 100, the threshold is at the value given by "x3". This parameter affects the speed of the hand movement in the following way: • EMG = "x2" → Flexion/extension speed is 1 • EMG = (EMG value at the threshold – 1) → Flexion/extension speed is 1023 • EMG = EMG value at the threshold → Pronation/supination speed is 1 • EMG ≥ "x3" → Pronation/supination speed is 1023 • The value of the speed linearly increases between the border values given above
"x5" description	This byte is the <i>"Rotation period threshold"</i>. For further information, see the section 5.1 in the document "Deliverable D1.2". It can be in the range of 1 to 10 (binary value). The default value is 1. A real period is obtained as follows: $period = (x5 \cdot 100) [ms]$.
"x6" description	This byte is the <i>"Co-contraction intensity threshold"</i>. For further information, see the section 5.1 in the document "Deliverable D1.2". It can be in the range of 1 to 100 (binary value). The default value is 50. By setting this parameter to 1, the threshold is at the 1% of the calculated EMG range ["x2", "x3"], while on the other hand, by setting this parameter to 100, the threshold is at the value given by "x3".
"x7" description	This byte is the <i>"Co-contraction period threshold"</i>. For further information, see the section 5.1 in the document "Deliverable D1.2". It can be in the range of 1 to 10 (binary value). The default value is 1. A real period is obtained as follows: $period = (x7 \cdot 100) [ms]$.

Table 3.8. Command 8 – Reset EMG calibration.

Command	> LEN E M G R CS <
Decimal view	62 5 69 77 71 82 48 60
Hex view	0x3E 0x05 0x45 0x4D 0x47 0x52 0x30 0x3C
Command length	8
Description	This command resets the calibrated EMG values on both channels to default ones [102, 613].

Table 3.9. Command 9 – Setting Michelangelo hand control algorithm.

Command	> W x <
Decimal view	62 87 x 60
Hex view	0x3E 0x57 REGIME 0x3C
Command length	4
Description	This command defines the chosen Michelangelo hand control algorithm according to the "x" byte. The "x" byte can be either the ASCII character "A" or the ASCII character "T". • "x" = "A" – AxonMaster hand control algorithm is the active one; • "x" = "T" – TR-FB hand control algorithm is the active one; For further information, see sections 5 and 7 in the document "Deliverable D1.2". If the command is received properly, the TR-FB device will respond with the message consisting of ASCII characters (">OK<").

Table 3.10. Command 10 – New pair.

Command	> B T x1 x2 x3 x4 x5 x6 <
Decimal view	62 66 84 x1 x2 x3 x4 x5 x6 60
Hex view	0x3E 0x42 0x54 x1 x2 x3 x4 x5 x6 0x3C
Command length	10
Description	This command requires a new hand to be connected to the TR-FB device. “x1”, “x2”, “x3”, “x4”, “x5”, “x6” are the bytes (binary values) that represent an address of a new hand to which the device has to connect.

Table 3.11. Command 11 – Feedback look-up tables activation.

Command	> T B x1 x2 <
Decimal view	62 84 66 x1 x2 60
Hex view	0x3E 0x54 0x42 x1 x2 0x3C
Command length	6
Description	This command defines which look-up table(s) will be used as feedback coding scheme(s). More precisely, it defines which signal(s) will be used as a feedback source (“x1” – binary value), as well as whether a special condition for each table will be used (“x2” – binary value). For further information, see the section 7 in the document “Deliverable D1.2”. If the command is received properly, the TR-FB device will respond with the message consisting of ASCII characters (“>OK<”).
“x1” description	This byte is in the range of 0 to 63 and is calculated as follows: $x1 = \text{sum}([TB1, TB2, TB3, TB4, TB5, TB6, 0, 0] \cdot [2^0, 2^1, 2^2, 2^3, 2^4, 2^5, 2^6, 2^7])$, where $TBn = \begin{cases} 0, & \text{if } TBn \text{ will not be used as a feedback coding scheme} \\ 1, & \text{if } TBn \text{ will be used as a feedback coding scheme} \end{cases}$ • TB1 – “Flexor” table • TB2 – “Extensor” table • TB3 – “Setpoint” table • TB4 – “Force” table • TB5 – “Aperture” table • TB6 – “Rotation” table
“x2” description	This byte is in the range of 0 to 63 and is calculated as follows: $x2 = \text{sum}([CTB1, CTB2, CTB3, CTB4, CTB5, CTB6, 0, 0] \cdot [2^0, 2^1, 2^2, 2^3, 2^4, 2^5, 2^6, 2^7])$, where $CTBn = \begin{cases} 0, & \text{if an appropriate condition will not be used} \\ 1, & \text{if an appropriate condition will be used} \end{cases}$ • TB1 – a special condition for the “Flexor” table • TB2 – a special condition for the “Extensor” table • TB3 – a special condition for the “Setpoint” table • TB4 – a special condition for the “Force” table • TB5 – a special condition for the “Aperture” table • TB6 – a special condition for the “Rotation” table

Table 3.12. Command 12 – Feedback look-up table.

Command	> T x x0_0L x0_0H x0_1L x0_1H ... x0_8L x0_8H x1_0L x1_0H x1_1L x1_1H ... x1_8L x1_8H x2_0L x2_0H x2_1L x2_1H ... x2_8L x2_8H x3_0L x3_0H x3_1L x3_1H ... x3_8L x3_8H x4_0L x4_0H x4_1L x4_1H ... x4_8L x4_8H
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	x5_0L x5_0H x5_1L x5_1H ... x5_8L x5_8H x6_0L x6_0H x6_1L x6_1H ... x6_8L x6_8H x7_0L x7_0H x7_1L x7_1H ... x7_8L x7_8H x8_0L x8_0H x8_1L x8_1H ... x8_8L x8_8H x9_0L x9_0H x9_1L x9_1H ... x9_8L x9_8H x10_0L x10_0H x10_1L x10_1H ... x10_8L x10_8H <
Decimal view	62 84 x x0_0L x0_0H x0_1L x0_1H ... x10_8L x10_8H 60
Hex view	0x3E 0x54 x x0_0L x0_0H x0_1L x0_1H ... x10_8L x10_8H 0x3C
Command length	184
Description	This command sends one feedback look-up table. For further information, see the section 7 in the document “Deliverable D1.2”. If the command is received properly, the TR-FB device will respond with the message consisting of ASCII characters (“>OK<”).
“x” description	This byte defines which look-up table is being sent. It can be in the range of 1 to 6 (binary value): • TB1 – “Flexor” table • TB2 – “Extensor” table • TB3 – “Setpoint” table • TB4 – “Force” table • TB5 – “Aperture” table • TB6 – “Rotation” table
“xn_0L” description	This byte defines the lower limit of the feedback source signal. It can be in the range of 0 to 100 (binary value).
“xn_0H” description	This byte defines the upper limit of the feedback source signal. It can be in the range of 0 to 100 (binary value).
“xn_mL” description	This byte defines the lower limit of the vibration intensity of the <i>m</i> -th motor. It can be in the range of 0 to 100 (binary value).
“xn_mH” description	This byte defines the upper limit of the vibration intensity of the <i>m</i> -th motor. It can be in the range of 0 to 100 (binary value).

Table 3.13. Command 13 – Real time clock.

Command	> R T C x1 x2 x3 x4 x5 x6 x7 <
Decimal view	62 82 84 67 x1 x2 x3 x4 x5 x6 x7 60
Hex view	0x3E 0x52 0x54 0x43 x1 x2 x3 x4 x5 x6 x7 0x3C
Command length	12
Description	This command sends a real time clock to the TR-FB device, where: • “x1” – hour (binary value) • “x2” – minute (binary value) • “x3” – second (binary value) • “x4” – day of the week (binary value) • “x5” – month (binary value) • “x6” – day of the month (binary value) • “x7” – (year – 1980) (binary value) If the command is received properly, the TR-FB device will respond with the message consisting of ASCII characters (“>OK<”).

Table 3.14. Command 14 – uSD logging.

Command	> S D x <
Decimal view	62 83 68 x 60
Hex view	0x3E 0x53 0x44 x 0x3C

Command length	5
Description	This command will either start or stop data logging on a uSD card, depending on the value of the “x” byte. The “x” byte can be either 1 (binary value) or 0 (binary value). By setting this byte to 1, the TR-FB device will start data logging. On the other hand, by setting this byte to 0, uSD data logging will be stopped. If the command is received properly, the TR-FB device will respond with the message consisting of ASCII characters (“>OK<”).

Table 3.15. Command 15 – EMG filter.

Command	> N A ; x <
Decimal view	62 78 65 59 ; 60
Hex view	0x3E 0x4E 0x41 0x3B x 0x3C
Command length	6
Description	This command sets the filter strength parameter according to the value of the “x” byte. This byte can be in the range of 1 to 255 (binary value). The default value is 20. For further information, see the section 3 in the document “Deliverable D1.2”. If the command is received properly, the TR-FB device will respond with the message consisting of ASCII characters (“>OK<”).

4. Commands from a PC terminal to the TR-FB device

The commands that can not be sent from the PC application, but the TR-FB device may receive them, are given in the tables below.

Table 4.1. Command 1 – ERM configuration.

Command	> E R M ; x1 x2 <
Decimal view	62 69 82 77 59 x1 x2 60
Hex view	0x3E 0x45 0x52 0x4D 3B x1 x2 0x3C
Command length	8
Description	This command initializes the vibromotor driver for ERM motors. “x1” and “x2” are two ASCII bytes that represent ERM motor voltage multiplied by 10. Maximal voltage that can be applied to motors is 5V. The TR-FB device will respond either with the ASCII message “>OK<” or “>ERR<”.
Example	>ERM;35< - Initialize the motor driver for ERM motors and set the output voltage to 3.5V.

Table 4.2. Command 2 – LRA configuration.

Command	> L R A ; x1 x2 ; x3 x4 x5 <
Decimal view	62 76 82 65 59 x1 x2 59 x3 x4 x5 60
Hex view	0x3E 0x4C 0x52 0x41 3B x1 x2 0x3B x3 x4 x5 0x3C
Command length	12
Description	This command initializes the vibromotor driver for LRA motors. “x1” and “x2” are two ASCII bytes that represent LRA motor voltage multiplied by 10. Maximal voltage that can be applied to motors is 5V. “x3”, “x4” and “x5” are three ASCII bytes that represent LRA motor frequency in Hz. The frequency can be set in the range of 125 Hz to 300 Hz. The TR-FB device will respond either with the ASCII message “>OK<” or “>ERR<”.

Example	>LRA;35;180< - Initialize the motor driver for LRA motors and set the output voltage to 3.5V and the frequency to 180 Hz.
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Table 4.3. Command 3 – Fuel gauge.

Command	> F G <
Decimal view	62 70 71 60
Hex view	0x3E 0x46 0x47 0x3C
Command length	4
Description	This command initializes fuel gauge parameters for battery monitoring. If the command is received properly, the TR-FB device will respond with the message consisting of ASCII characters (">OK<").

5. Data stream package (data from the TR-FB device to the PC application)

Once the start data stream command (*Command 1*) is sent to the TR-FB device, it will start streaming data packages in the form as it is given in *Table 5.1*. Similarly to the third section, bytes “LEN” and “CS” are calculated in the following way:

$$LEN = Package\ length - 3$$

$$CS = rem(sum(LEN, "S", "T", B1, B2, ..., Bn), 256),$$

where:

- B1, B2, ... Bn are the data streaming items and are given in *Table 5.2*.

For further information, see the section 6 in the document “Deliverable D1.2”.

Table 5.1. Data stream package.

Data stream package	> LEN S T B1 B2 ... Bn CS <
Decimal view	62 33 83 84 B1 B2 ... Bn CS 60
Hex view	0x3E 0x21 0x53 0x54 B1 B2 ... Bn CS 0x3C
Package length	36

Table 5.2. Data stream items.

Byte number	Item name	Item source	Item range (binary values)
B1	Hand connection status	TR-FB	0 – hand not connected 1 – hand connected
B2	EMG CH1 low byte	DIAC0 ¹	0 – 255
B3	EMG CH1 high byte		0 – 3
B4	EMG CH2 low byte	DIAC1	0 – 255
B5	EMG CH2 high byte		0 – 3
B6	Hand strength (set-point) low byte	DIAC11	0 – 255
B7	Hand strength (set-point) high byte		0 – 3
B8	Hand battery status	DIAC13	0 – 100

¹ DIAC - Data stream item activation code – according to Ottobock protocol (Michelangelo hand is the item source).

B9	Freecounter	DIAC 39	0 – 255
B10	Main set-point position	DIAC 72	0 – 255
B11	Main actual position	DIAC 75	0 – 255
B12	Hand mech grip state	DIAC 82	0 – lateral mode 1 – opposition mode 4 – neutral mode
B13	Hand motion type	DIAC 84	0 – lateral close 1 – opposition close 2 – opposition open 3 – lateral open
B14	Thumb force	DIAC 87	0 – 255
B15	Rotator set-point position	DIAC 104	0 – 255
B16	Rotator actual position	DIAC 105	0 – 255
B17	TR-FB battery status	TR-FB	0 – 100
B18 ²	Vibration motors statuses	TR-FB	0 – 255
B19	Vibration intensity of the motor 1	TR-FB	0 – 100
B20	Vibration intensity of the motor 2	TR-FB	0 – 100
B21	Vibration intensity of the motor 3	TR-FB	0 – 100
B22	Vibration intensity of the motor 4	TR-FB	0 – 100
B23	Vibration intensity of the motor 5	TR-FB	0 – 100
B24	Vibration intensity of the motor 6	TR-FB	0 – 100
B25	Vibration intensity of the motor 7	TR-FB	0 – 100
B26	Vibration intensity of the motor 8	TR-FB	0 – 100
B27	Hand speed TRFB-Control low byte	TR-FB	0 – 255
B28	Hand speed TRFB-Control high byte	TR-FB	0 – 3
B29	Normalized EMG CH1	TR-FB	0 – 100
B30	Normalized EMG CH2	TR-FB	0 – 100

² B18 is calculated as follows:

$$B18 = \text{sum}([VB1, VB2, VB3, VB4, VB5, VB6, VB7, VB8] .* [2^0, 2^1, 2^2, 2^3, 2^4, 2^5, 2^6, 2^7]),$$

where $VBn = \begin{cases} 0, & \text{if } VBn \text{ is turned off} \\ 1, & \text{if } VBn \text{ is turned on} \end{cases}$